Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/23/2008
PI Name:	Miller, Jack Ph.D.		
Project Title:	Measurements of fragmentation cross sections and particle spectra for galactic cosmic ray-like nuclei between 3 and 10 GeV/nucleon at the BNL AGS		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	miller@lbl.gov	**Fax:**	FY 510 486-6949
PI Organization Type:	GOVERNMENT	**Phone:**	510 486-7130
Organization Name:	Ernest O. Lawrence Berkeley National Laboratory		
PI Address 1:	Mail Stop 74-197, Life Sciences Division		
PI Address 2:	1 Cyclotron Road		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	03/24/2006	**End Date:**	03/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2008, per J. Dardano/JSC (9/08) NOTE: Received NCE to 9/30/2007 per P. Denkins/JSC (5/07) NOTE: Corrected start/end dates to 3/24/2006-6/30/2007 per P. Denkins/JSC (5/07); previously had end date of 3/1/2008		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heilbronn, Lawrence (Lawrence Berkeley National Laboratory) Zeitlin, Cary (Lawrence Berkeley National Laboratory) Adams, James (NASA-Marshall Space Flight Center) Guetersloh, Stephen (Lawrence Berkeley National Laboratory) Pinsky, Lawrence (University of Houston)		
Grant/Contract No.:	NNJ06HD46I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is for accelerator measurements of the products of the interactions of energetic protons and heavy ions in elemental targets and radiation shielding materials. Two types of measurements will be made: (1) fragmentation cross sections, or interaction probabilities in thin elemental targets and (2) fluences of fragments produced by interactions in thick elemental and composite targets. The objective is to improve the accuracy of nuclear transport models for space radiation applications. These measurements will be used to improve the cross section database used in these models both directly and through improvements in nuclear fragmentation models. The goal is obtain transport models that are accurate to ±25% for the fluxes of the major components emerging from the shielding. Comparing measured fluences with model calculations will test this accuracy. The range of projectile atomic numbers for these measurements will be 1-26. The range of projectile energies will be 100-2000 MeV/nucleon. Differential cross sections for production of charged nuclear fragments, neutrons and (if deemed necessary) pions will be measured as a function of one or more observables including projectile energy, projectile mass, fragment charge, fragment energy and emission angle, according to the requirements defined by the model calculations. Projectiles, projectile energies, targets and other parameters will be selected in consultation with the developers of the transport models. The detectors to be used will include individual and position-sensitive silicon detectors, silicon strip detectors and highly pixilated arrays. The latter two will be used to help separate out multiple fragments, thereby improving the accuracy of the cross sections for light nuclei. Liquid and solid plastic scintillators will be used to measure neutron energy spectra. The accelerator facilities to be used will be the Brookhaven National Laboratory AGS and Booster Application Facility (BAF), the National Institute of Radiological Sciences Heavy Ion Medical Accelerator at Chiba (HIMAC), and perhaps others, such as the proton synchrotron at the Loma Linda University Medical Center and the National Superconducting Cyclotron Laboratory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	None.
Task Progress:	Data analysis has been completed. Systematics of fragment production cross sections as a function of projectile and target mass and energy and fragment charge have been established and compared to previous data and to fragmentation models. Data are available on websites at Lawrence Berkeley National Laboratory and The University of Houston. Data will be transferred to a server at NASA-LaRC. http://fragserver.lbl.gov/ (Ed. note 5/26/16--website no longer accessible.)
Bibliography Type:	Description: (Last Updated: 01/21/2020)